Conjugated Systems & Aromaticity

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Contents

1 Aromaticity & Aromatic Substitution

1.1 Nucleophilic Aromatic Substitution

Core Concept 1.1: Nucleophilic Aromatic Substitution

Nucleophilic aromatic substitution (NAS or S_NAr) is an interaction in which the aromatic ring is the electrophile. NAS comes in two "flavors": **diazonium** (*Core Concept 1.2*) or **addition/elimination** (*Core Concept 1.3*) chemistry.

*Recall that in electrophilic aromatic substitution (**??**), the aromatic ring is the nucleophile.*

Core Concept 1.2: Diazonium Chemistry

Diazonium chemistry begins by N_2 leaving a phenyldiazonium molecule—via breaking the bond between the σ C(sp^2)–N(sp) bond—to produce a carbocation possessing benzene ring and an extermely stable N_2 gas moleucle.

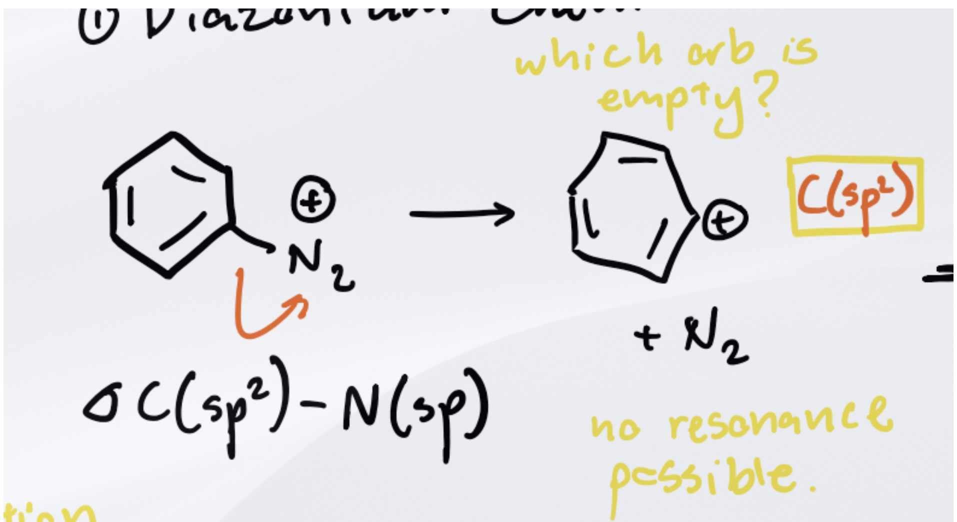

Figure 1

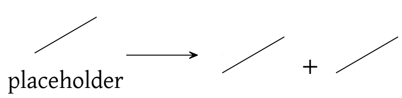

Figure 2: caption

It's important to notice that resonance is not possible in this specific carbocation state because the orbtial that is empty is the C(sp^2) orbtial. We would need an unbalanced p orbital for resonance to occur. As a result, this phenyl cation is highly reactive and hungry for any nearby nucleophile to help stabilize its positive charge.

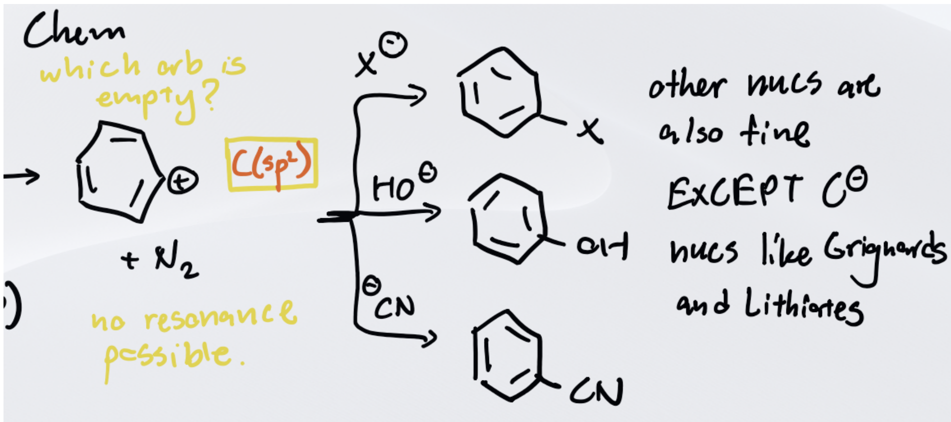

Figure 3

figureouthowtodomultiplearrows

Figure 4: caption

As shown above, nearly all covered nucleophiles work in this interaction **with the exception of grignards and lithiates**. *The reasoning behind this will not be covered in the scope of these notes.*

Combining this diazonium chemistry with previously learned reactions, chemists can effectively attach nearly any nucleophile to a benzene molecule using the following reaction pathway.

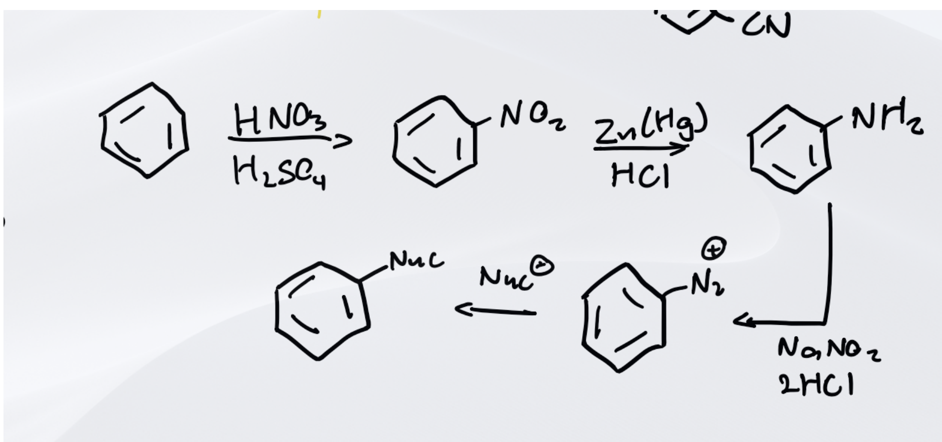

Figure 5: reference the core concepts for the reactions within the caption

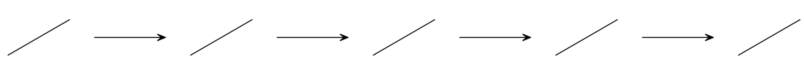

Figure 6: caption

Core Concept 1.3: Addition/Elimination

The NAS **addition/elimination** reaction occurs in the following general manner.

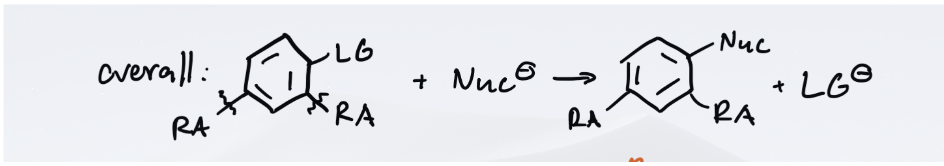

Figure 7

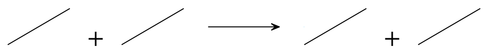

Figure 8: caption

Recall that we have seen resonance acceptors in resonance M directors (**??**), however, the these resonance acceptors are not for the purpose of directing meta in the context of NAS chemistry. Their main purpose is to be able to accept e$^-$ density after the nucleophile attack the carbon with the leaving group.

Additionally, this reaction requires a strong nucleophiles because it ultimately needs to break aromaticity as shown in the mechanism below. It should be noted that this step—the addition—is consequently the rate determining step.

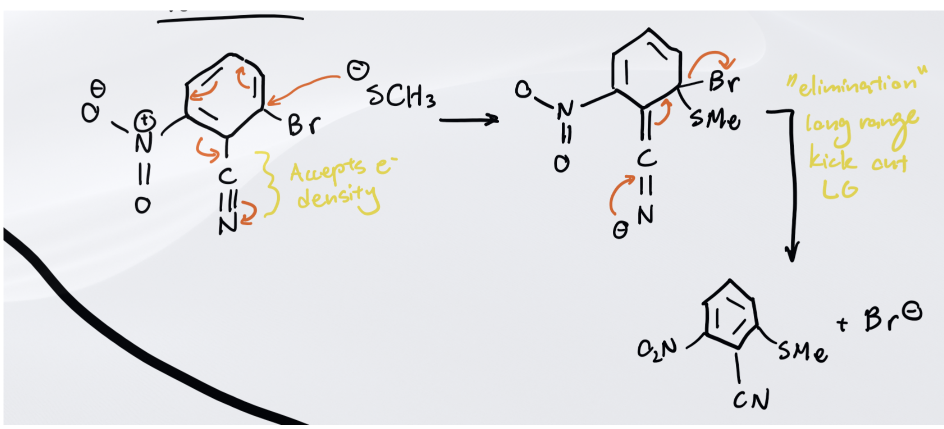

Figure 9

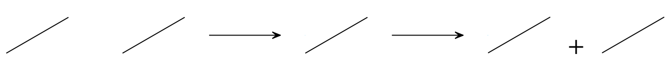

Figure 10: caption

As the mechanism showcases, the reaction is simply a relay of e$^-$ density to the resonance acceptor, which then bounces back the e$^-$ density in order to kick out the leaving group in the end. However, since the outcome of this interaction is ultimately the same as that of an S_N2 reaction, one may wonder why S_N2 is not a viable pathway.

Revisiting the S_N2 mechanism, recall that, in the lense of molecular orbitals, the nucleophile interacts with the σ^* anti-bonding orbital between the electrophile and leaving group in order to kick out the leaving group and create a bond simultaneously via a backside attack.

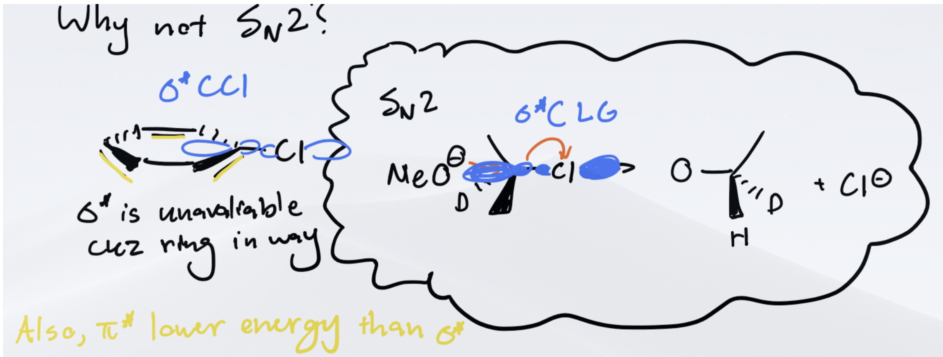

Figure 11

Figure 12: caption

In the aromatic ring scenario, this nessecary σ^* interaction is extremely unlikely due to the physcial barrier to entry because of the ring itself.

In the context of the frontier molecular orbital theory, the π^* orbital is lower in energy than σ^*, making it the LUMO and a much more accessible orbital to interact with, which is what happens in addition/elimination, the favorable pathway.

Lets go through two examples.

(a) **Example 1: Predicting Major Kinetic Product**

Consider the following reaction consisting to two possible leaving groups F and Cl with a NO_2 resonance acceptor.

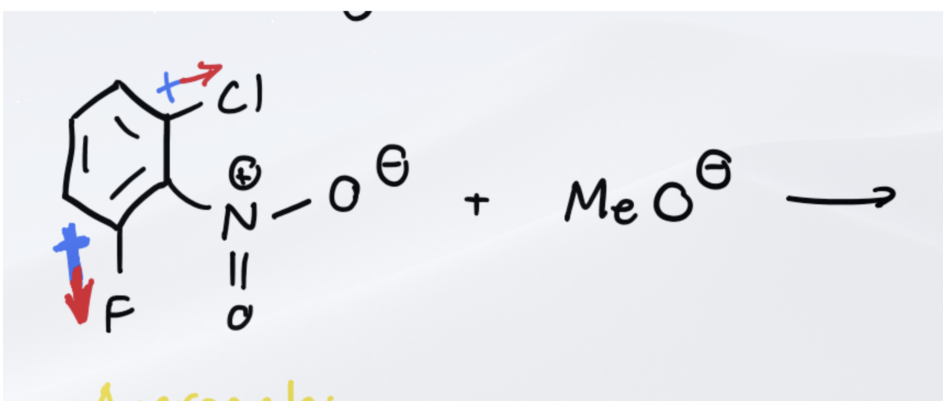

Figure 13

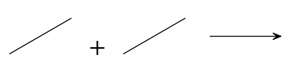

Figure 14: caption

Both addition/elimination pathways are possible as shown below, ultimately leading to two distinct products.

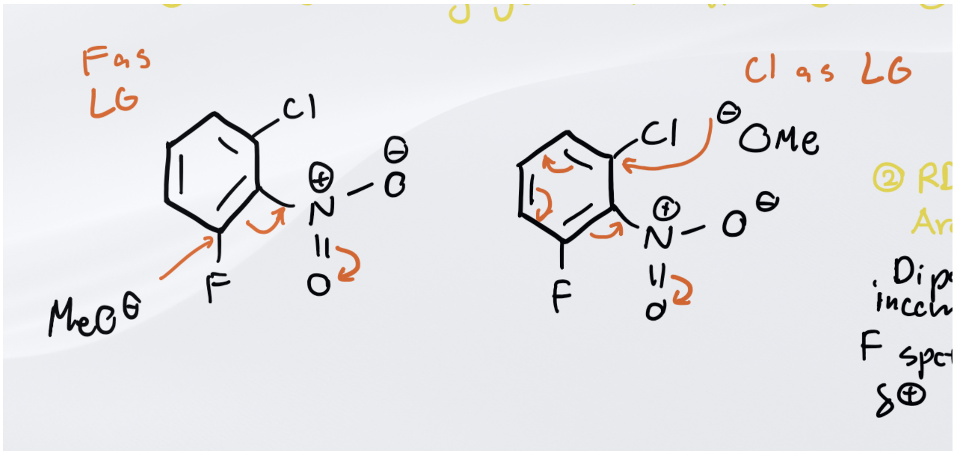

Figure 15

Figure 16: caption

Since the nucleophile is held constant across both reaction pathways, the ability of the nucleophile to break aromaticity is not a factor. Instead, it is the greater dipole moment created by the more electronegative fluorine over the chlorine. Due to the greater partial positive charge of the fluorine carbon, MeO^- favors attacking that carbon because it its more electrophilic. Thus the reaction favors the substitution of the F over the Cl.

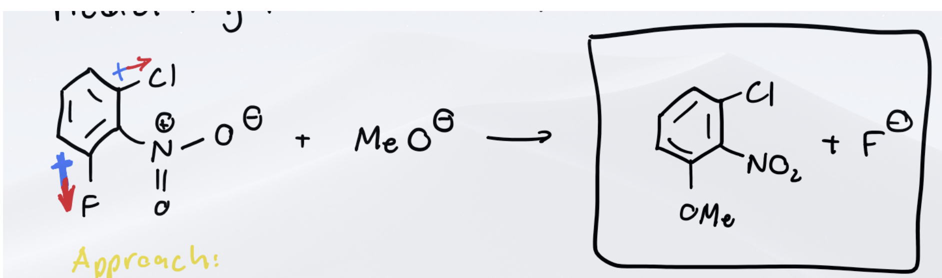

Figure 17

Figure 18: caption

(b) **Example 2: Predicting Product of NAS Chemistry**

Consider the following addition/elimination reaction below.

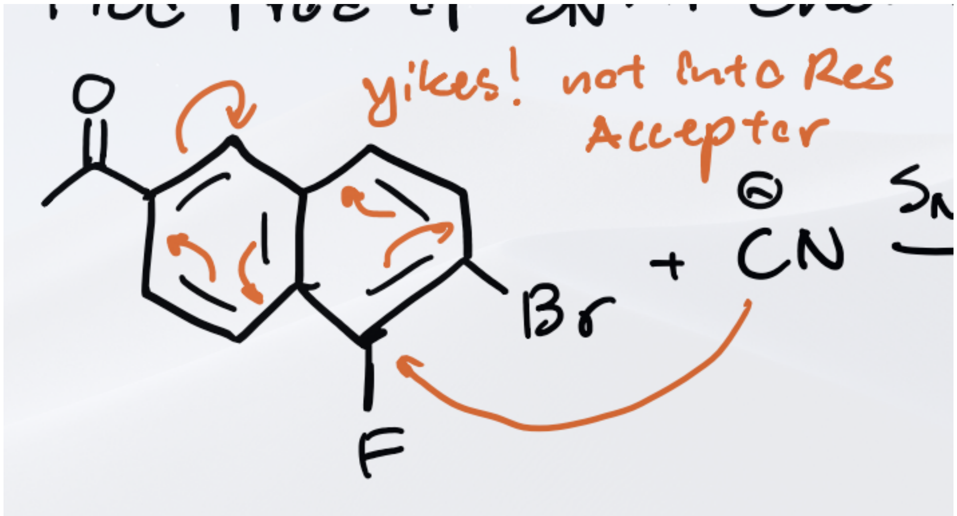

Figure 19: short initial reactants without arrows

Figure 20: caption

In this case, simply only one of the leaving group substitutions work to donate e$^-$ density into the resonance acceptor and the other doesn't.

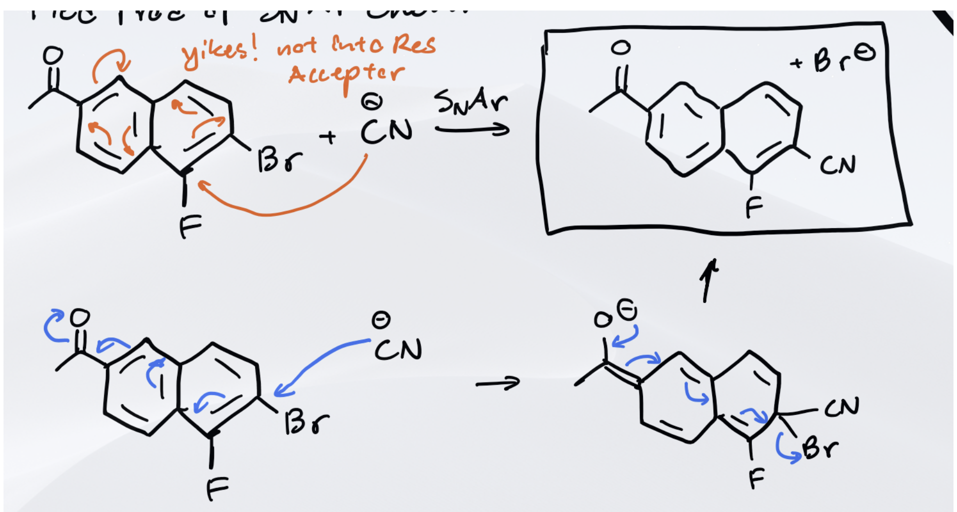

Figure 21: showcase the pathways and how one isnt possible

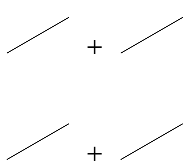

Figure 22: caption

Thus the exclusive reaction outcome maps out as follows.

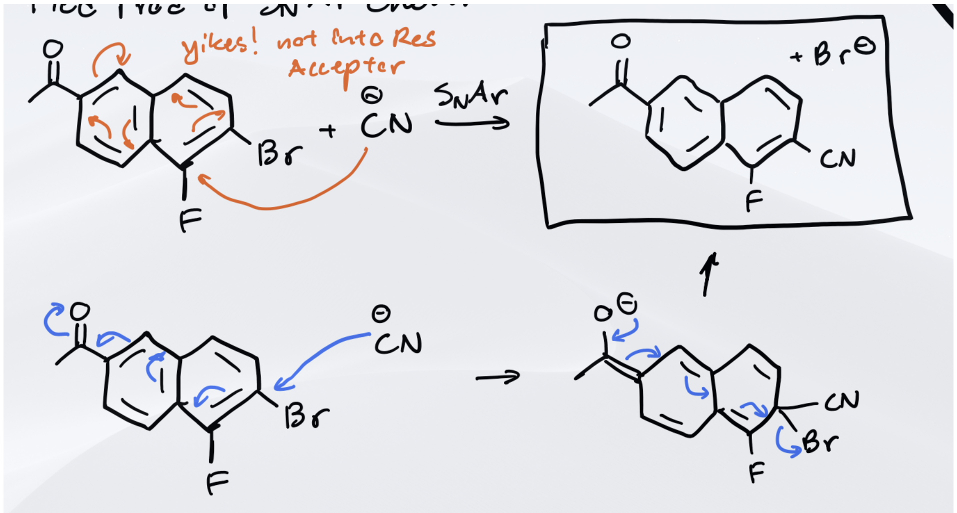

Figure 23: complete chemical equation

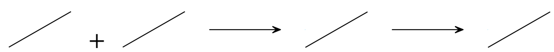

Figure 24: caption

(c) **Example 3: Predicting Starting Material**

Consider having to determine the starting aromatic compound to the following products based on the given nucleophile.

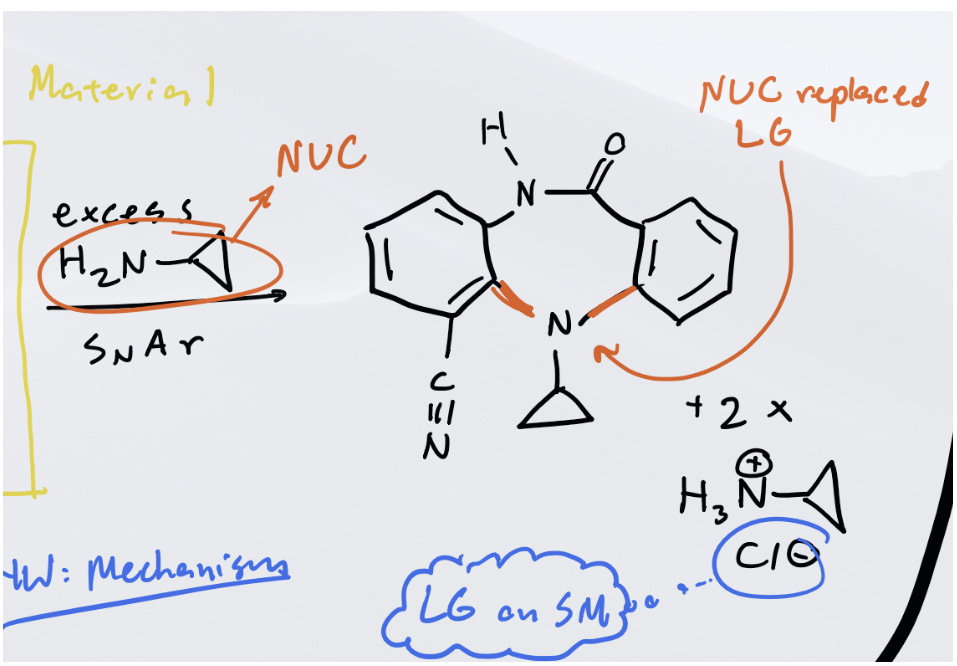

Figure 25

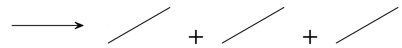

Figure 26: caption

Because we see two Cl^- and two H_3N^+R on the product side, we know that the addition/elimination has occured twice. Luckily the product quite literally has the nucleophile placed directly on center connected to the aromatic ring system via two bonds on the nitrogen. Based on this, we can reasonably predict the starting material to have had these two bond substituted.

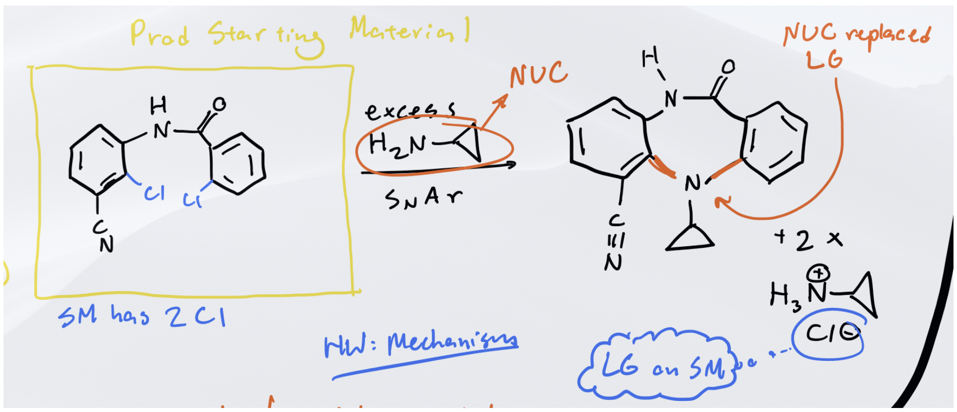

Figure 27

Figure 28: caption